

End-to-end experimental methodology

Data preparation



Ground PM2.5 + meteorology
Delhi 40 · Kolkata 10 · Guwahati 4
(3-hourly)

Pre-processing & imputation

- feature engineering (F = 14: PM2.5, AT, RH, WS, WD sin/cos, cyclic time, season)
- causal gap-filling
- interleaved 70/15/15 split
- per-(month, hour) climatology-residual target

Cleaned per-city
PM2.5 tensors

Cleaned meteorology

Input data

Modelling & cross-city transfer

multi-step
forecast

LSTM-TL
(station-independent)
pre-train → fine-tune

Inductive ST-GNN
Stage 0 pre-train →
Stage 1 PT-FT / Stage 2 Graph-DANN

k-NN graph construction
(k = 3, Gaussian edge weights)

Three-way verification
zero-shot · scratch · transfer
(real iff transfer > both)

Evaluation on target test
 R^2 · MAE · RMSE · MAPE (raw $\mu\text{g}/\text{m}^3$)

